IIT DELHI
MASTER IN TECHNOLOGY, ROBOTICS
JRD 891: MTECH Minor Project Presentation
Visual SLAM for Digital Heritage
Guide: Prof. Chetan Arora
Presented By: 2024JRB2028 N Parthasaradhi Reddy
2024JRB2066 C Adithya
Date: 08.05.2025

Problem Statement

Challenges in Archaeology:

- **Fragility of Heritage Sites:** Physical excavation can cause irreversible damage to artifacts and historical layers, leading to permanent loss.
- **Loss of Contextual Data:** Without systematic tracking, crucial spatial and structural information may be lost during excavation.

Proposed Solution

- **Digital Reconstruction System**
  - Create accurate virtual representation of archaeological sites
  - Ensure data preservation without physical risks
- Develop portable device for efficient recording of sites and objects

Key Features:

- **Real-time 3D Mapping:** Capture high-resolution spatial data
- **Version Control:** Maintain chronological 3D digital twins
- **Digital Backup:** Preserve data against physical degradation

Methodology

1. Identify SLAM requirements
2. Implement SLAM for Mapping
3. Develop ORB SLAM application (Modular)
4. Photo Slam Integration
5. Evaluate Performance in Various Environments
6. Identify & Address Issues

Progress up to Mid Term

- Developed Python application with two modes:
  - User Mode: Live video and live map
  - Debug Mode: Additional features like detected feature points
- Map naming prompt before saving

ORB Slam vs Photo Slam

ORB-SLAM

- Feature-based SLAM using ORB key points
- Reconstructs sparse 3D point cloud — only (x, y, z) positions
- Fast and efficient; suitable for real-time navigation
- No color, surface detail, or realism

Photo SLAM

- Dense SLAM technique for photo-realistic reconstruction
- Models scene using 3D Gaussians — (x, y, z, color, opacity, scale)
- Gaussians trained to fit scene features via photometric loss
- Enables realistic rendering; higher compute required

Viewers

Viewer	Status
GS Viewer	• Scene loads successfully • Random initial camera position
SIBR	• Compatibility issues found • Resolution underway
Photo SLAM	• Successful integration • Starts from initial frame

Table 1: Viewer Integration Status

Application Features

- ORB-SLAM integrated with:
  - New map creation
  - Feature detection
  - Save/Delete maps with names
  - Map loading
- Photo SLAM integrated
- Technique selection via dropdown menu
- Auto-viewer selection based on file type
- Added debug logging

Progress and Challenges

- Built custom Python GTK application
- Integrated both ORB-SLAM and Photo SLAM
- Implemented SLAM selection dropdown
- Developed viewer handling system
- Added logging for troubleshooting
- Auto-viewer launching for loaded maps

Results

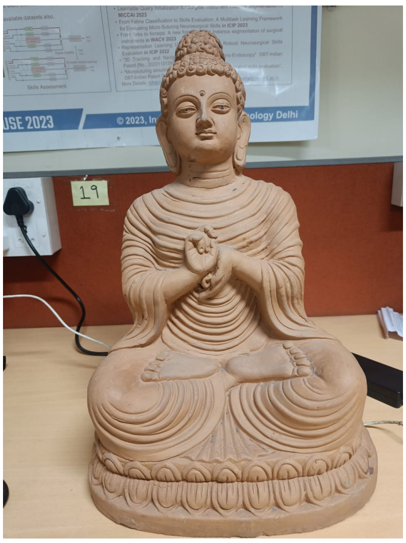

Figure 1: Artifact for Reconstruction

Figure 2: Reconstruted Image

Contributions

Frontend Development:

- Developed multiple interface versions
- Implemented core functionalities (Start, Stop, Features)
- Attempted embedded viewer integration

SLAM System:

- Installed and debugged ORB-SLAM and Photo SLAM
- Tested various viewer configurations

Backend Development:

- Implemented logging and map loading
- Debugged viewer performance issues
- Configured camera parameters
- Evaluated reconstruction techniques
- Debugged the Real Time Map Creation in Photo SLAM